

System Dynamics: Part III

ORSA Study text



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Sources:

MARTIN, Juan. *Vensim Online Course*. Online Course. 2024.

OpenAI. (2024). ChatGPT (Month Day version) [Large language model]. Available: <https://chat.openai.com>

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System Dynamics

INTRODUCTION

In this text, readers will learn follow-up information on how to work with Stock and Flow Diagrams with the VENSIM PLE program. Specifically, 1 solved example is presented here (Water tank).

What is learned in this exercise:

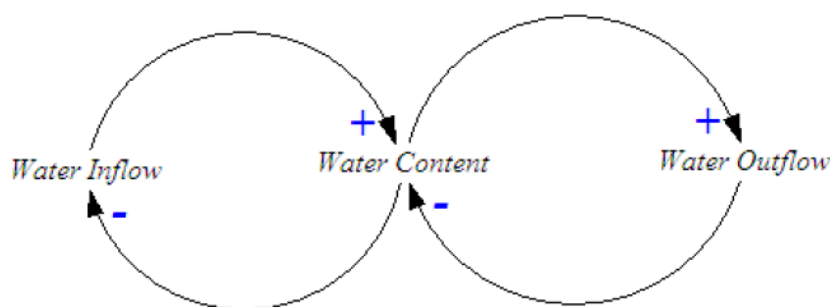
- To translate a Causal Loop Diagram CLD to a SFD
- Define the units of each variable and check them.
- Use Sim Setup simulation
- Manage, hide and compare simulations.

WATER TANK (EXAMPLE)

We want to simulate the behaviour of a tank containing liquid, which contains only one entry and one exit, both of which are always open. The tank has a capacity of 100 litres, and initially contains 50 litres of liquid.

We want to know the dynamic of the tank's content before making changes in the entry and exit flows. In other words, we want to make sure that the tank will not get overfilled and that it won't get empty.

The entry is regulated in a way that, to avoid overfilling, the flow increases when the tank is getting emptier, and the flow decreases when the tank gets filled up. Initially we equilibrate the entry in the tank in a way that enters a fraction of 1/10 of the empty volume in the tank.

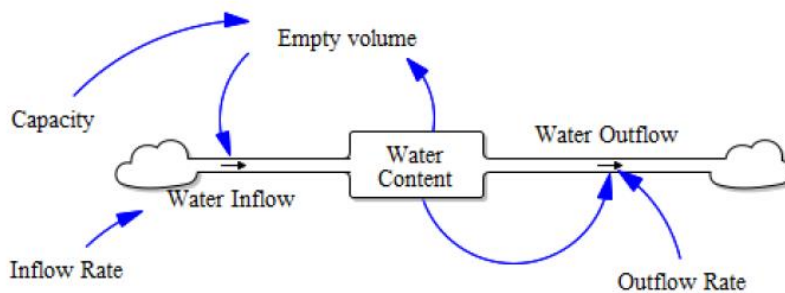


On the other hand, we regulate the exit in a way that, to avoid emptying, we will increase the exit flow when the tank is almost filled, and it will decrease when the tank becomes empty. Initially, we regulate the exit in a way that exits a fraction of 1/10 of the volume in the tank.

Questions:

1. What behavior would you expect, if initially, as we described earlier, the tank gets filled up at a rate of $1/10$ of its empty volume, and gets empty at the same rate?
2. What would happen if initially, the tank contains 80 litres?
3. What would happen if the entry is regulated at $1/20$ of the empty volume of the tank, and the exit at $1/10$ of the volume?

We want to create a model to understand the dynamics of the content of the tank to be able to answer those questions.



Model - Settings:



Initial time = 0 Final time = 100

Time step = 1 Units for time=Hour

Equations



(01) Capacity = 100

Units: Litres (write "Litres" because it is not on the menu)

(02) Empty volume = Capacity - Water Content

Units: Litres

(03) Inflow Rate = 0.1

Units: 1/Hour

The tank is filled every hour a 10% of its empty volume.

(04) Outflow Rate = 0.1

Units: 1/Hour

Every hour the tank is empty out a 10% of its contents.

(05) Water Content = Water Inflow - Water Outflow

Initial value: 50

Units: Litres

Variable Information	
Name	Water Content
Type	Level v Sub-Type v
Units	Liters v Check Units ☐ Supplementary
Group	v Min Max
Equations	Water Inflow-Water Outflow = INTEG (
Initial Value	50

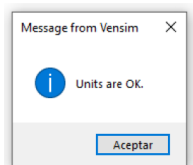
(06) Water Inflow= Empty volume*Inflow Rate

Units: Litres/Hour

(07) Water Outflow=Water Content*Outflow Rate

Units: Litres/Hour

Click the icon Units Check. You should see the message:



Now, simulate the model and make the next simulations.

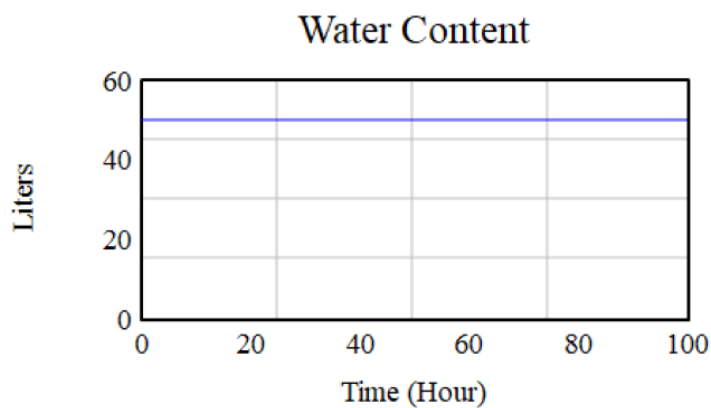


There are following three simulations. Please, answer the questions, change conditions and create a graph.

Simulation 1.

What behavior would you expect, if initially, as we described earlier, the tanks gets filled up at a rate of 1/10 of it empty volume, and gets empty at the same rate?

The water content would be constant at 50 liters and water inflow and outflow would also be constant at 5 liters.

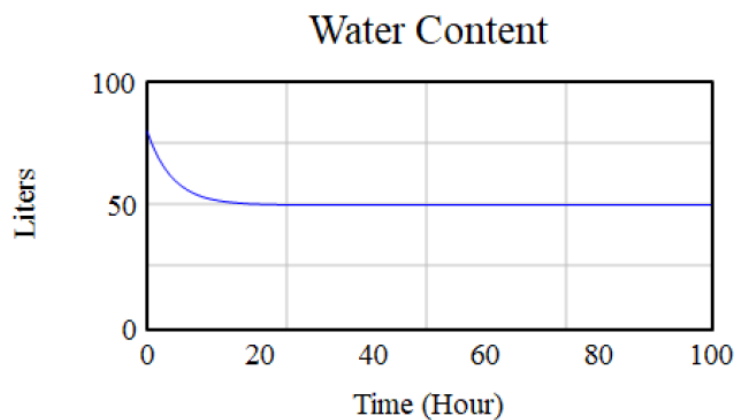


Simulation 2.

What would happen if initially, the tank contains 80 liters?

Change the initial value of the Stock and Simulate. Click Yes to override.

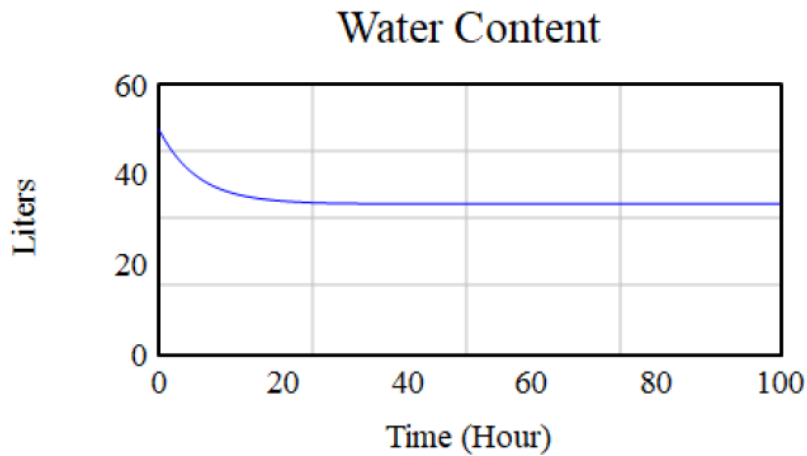
During the first period the inflow rate is $(100-80)/10=2$ liters, and the outflow rate is $80/10=8$ liters, this make that the contents drops until it reach the value of 50 liters, at this moment the inflow and outflow rates are $50/10=5$ and the water content is stable.



Simulation 3.

Change the initial value of the Stock to its initial value of 50. What would happen if the entry is regulated at 0.05 of the empty volume of the tank, and the exit at 0.10 of their content?

The water content at the first will fall down rapidly until 20 hours, after which it remain fairly constant at 33 litres. The water inflow will increase and the outflow will decrease at the beginning, and then remain constant at about 3 litres.

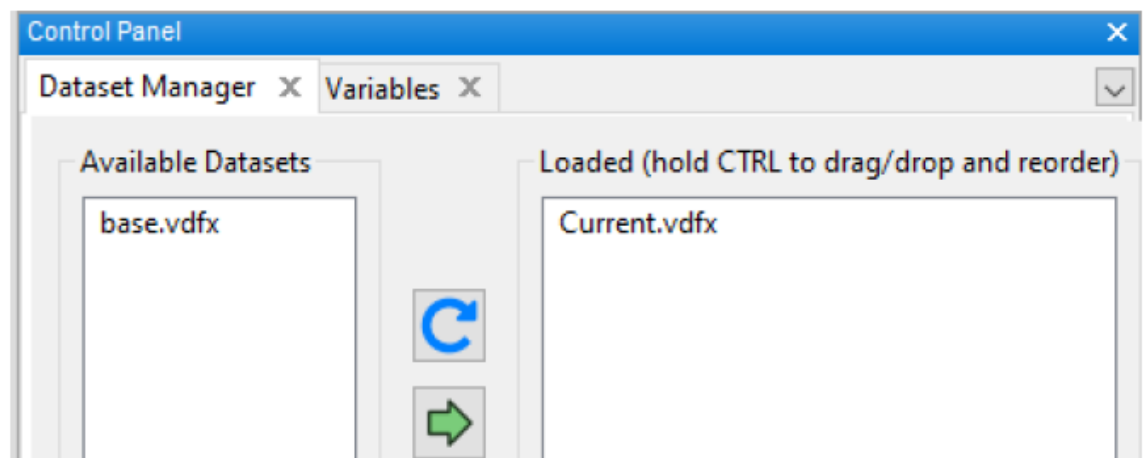


Notes:

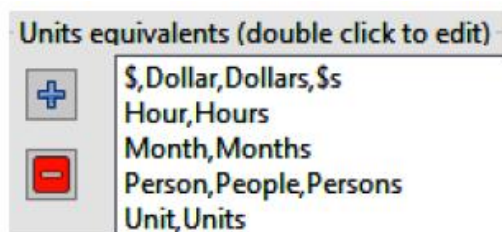
1. When you run a simulation you can save it as a dataset. These datasets can be managed from the Control Panel.



The default name for a simulation dataset is Current.vdfox.



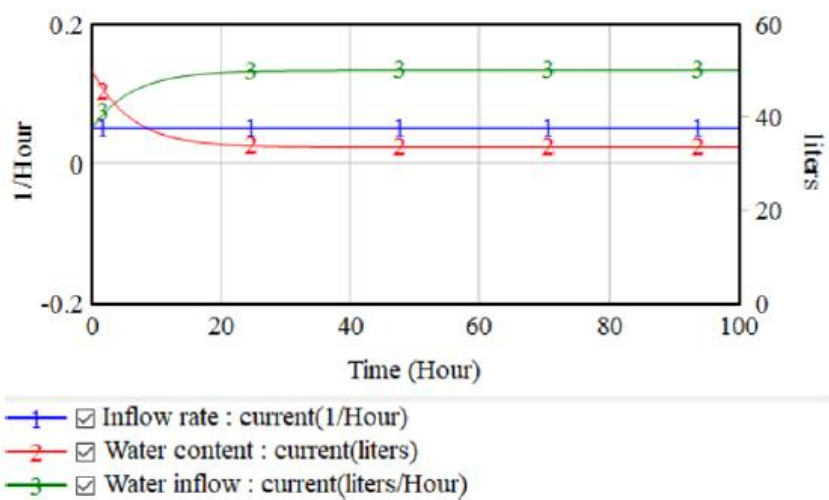
2. Go to Tools - Options - Units if you like to define synonyms like kilos, Kg, kg



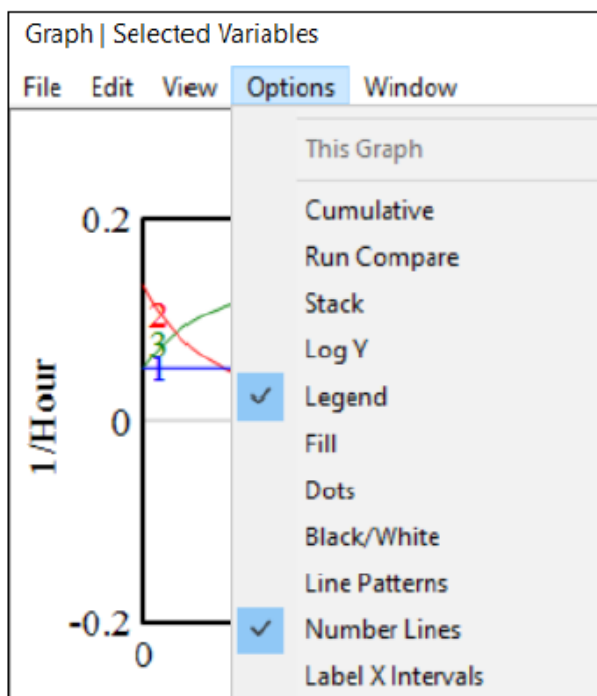
3. To display several variables on a single graph, follow these steps:

1. Hold down the Shift key.
2. Select one by one the variables.

3. Release the Shift key.
4. Click on the graph icon to generate the graph.



4. To add numbers to the lines select Options-Number Lines



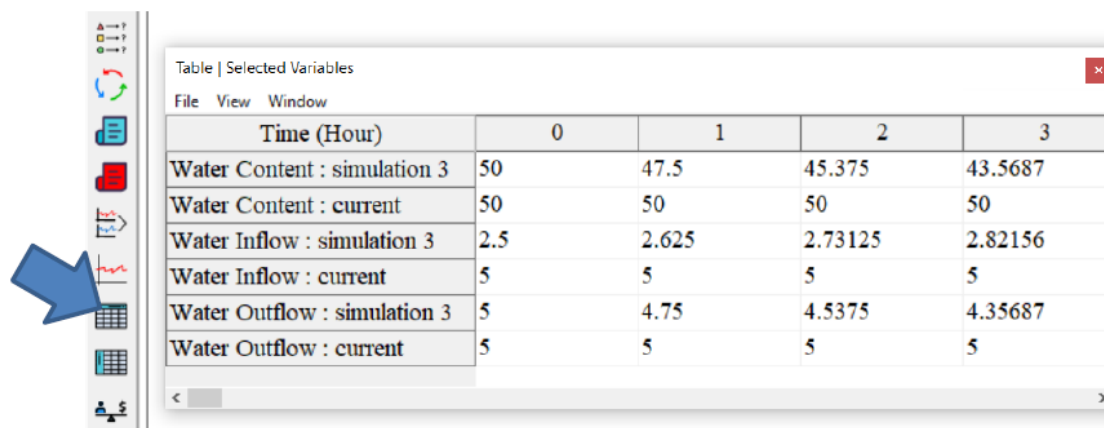
ANNEX 1 - COMPEARE SIMULATIONS in VENSIM

Click the Shift button and select the variables Water Content, Water Inflow, and Water Outflow, then click the Table icon to view their values.

Remember that the water inflow depends on the empty volume, and the water outflow depends on the water content.

In this simulation the initial water inflow is $(100-50)*0.05=2.5$ and the initial water outflow is $50*0.10=5$ this produces that the content of water decreases.

So, the water inflow increases because the empty volume increases $(100-47.5=52.5)$ and the water outflow decreases because the content decreases.



Time (Hour)	0	1	2	3
Water Content : simulation 3	50	47.5	45.375	43.5687
Water Content : current	50	50	50	50
Water Inflow : simulation 3	2.5	2.625	2.73125	2.82156
Water Inflow : current	5	5	5	5
Water Outflow : simulation 3	5	4.75	4.5375	4.35687
Water Outflow : current	5	5	5	5

ANNEX 2 – TRANSFER CLD TO SFD

Transferring a casual loop diagram to a stock and flow diagram involves several steps to ensure that the relationships and dynamics represented in the causal loop are accurately depicted in the stocks (quantities that accumulate over time) and flows (rates of change). Here's a step-by-step guide:

1. ***Identify Variables*:** Start by identifying all the key variables in your causal loop diagram. These include both the reinforcing and balancing loops.
2. ***Determine Stocks and Flows*:**
 - ***Stocks*:** Identify which variables represent accumulations (stocks). These are quantities that can increase or decrease over time (e.g., population, capital, inventory).
 - ***Flows*:** Determine the flows that affect these stocks. Flows represent changes in stocks over time (e.g., birth rate, investment rate, production rate).

3. ***Define Relationships***: Map out how the stocks and flows interact. For each stock, identify:

- Inflows (positive contributions)
- Outflows (negative contributions)

4. ***Create Flow Equations***: Develop mathematical relationships for each flow based on the causal relationships from the causal loop diagram. This can include rates that depend on various stocks or other flows.

5. ***Draw the Stock and Flow Diagram***:

- Use rectangles to represent stocks.
- Use arrows to represent flows (indicating the direction of change).
- Label each stock and flow clearly.
- Incorporate feedback loops to reflect the reinforcing and balancing nature from your causal loop diagram.

6. ***Validate the Model***: Ensure that the stock and flow diagram accurately represents the dynamics of the original causal loop diagram. Check for consistency in the relationships and that the feedback mechanisms are preserved.

By following these steps, you can effectively transition from a causal loop diagram to a stock and flow diagram, creating a comprehensive model that illustrates the system dynamics.

ANNEX 3 – SFD IN PRACTICE

Stock and flow diagrams are widely used across various fields due to their ability to model dynamic systems and illustrate how different variables interact over time. Here are some common applications in different domains:

1. Economics:

- Modeling national economies, including stocks like GDP, capital, and labor force, and flows such as investment, consumption, and taxation.
- Analyzing economic policies and their impacts on inflation, unemployment, and economic growth.

2. Environmental Science:

- Assessing natural resource management, such as water resources, where stocks represent water levels and flows indicate rates of consumption and replenishment.
- Modeling ecological systems, including population dynamics of species and nutrient cycling.

3. Public Health:

- Tracking disease spread, where stocks represent the number of infected individuals and flows represent transmission rates, recovery rates, and vaccination effects.

- Evaluating healthcare resources, such as hospital beds or medical supplies, and their utilization rates.

4. Business and Management:

- Inventory management, where stocks represent inventory levels and flows indicate production rates, sales rates, and restocking processes.
- Analyzing customer relationships, with stocks representing customer bases and flows indicating acquisition and churn rates.

5. Education:

- Understanding student enrollment and graduation rates, where stocks represent student populations and flows indicate admissions, graduations, and dropouts.
- Evaluating resource allocation in educational institutions, including teacher availability and student-teacher ratios.

6. Engineering and Manufacturing:

- Modeling production processes, where stocks represent work-in-progress inventories and flows indicate production rates and lead times.
- Analyzing supply chain dynamics, including stocks of raw materials and finished goods, and their respective inflow and outflow rates.

7. Social Sciences:

- Studying population dynamics, where stocks represent different demographic groups and flows indicate birth rates, death rates, and migration.
- Analyzing social behaviors and trends, such as the adoption of new technologies or social movements.

These applications highlight the versatility of stock and flow diagrams in providing insights into complex systems and facilitating informed decision-making.